

# Nikos Pagonas

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**Education** **Columbia University** Sep 2024 – Present  
*PhD in Computer Science* New York, NY, US  
Advisor: Prof. [Kostis Kaffes](#)  
GPA: 3.94/4.00  
Member of [DAPLab](#), focusing on improving the performance and efficiency of agentic serving. I build systems that exploit workflow structure and runtime characteristics to optimize model routing, scheduling, and resource allocation.

**National Technical University of Athens (NTUA)** Oct 2018 – Jun 2024  
*Master of Electrical and Computer Engineering* Athens, Greece  
Advisors: Prof. [Nikos Vasilakis](#), Prof. [Georgios Goumas](#)  
GPA: 9.39/10.00 (14<sup>th</sup> out of 344 students)  
Thesis: “*SPLASH: Scaling Out Shell Scripts on Serverless Platforms*”

**Experience** **Microsoft Research** May 2026 – Aug 2026  
*Research Intern* Redmond, WA, US  
Supervisor: [Sharad Agarwal](#)  
Analyzed production traces to motivate latency awareness in the [Microsoft Foundry model router](#). Worked on adding model-routing support to an LLM serving simulator used by Azure CoreAI, aiming for higher-fidelity opportunity analysis through production-trace replay.

**Google** May 2025 – Aug 2025  
*Student Researcher* Sunnyvale, CA, US  
Supervisor: Prof. [Arvind Krishnamurthy](#)  
Designed and implemented [Cortex](#), a prototype agentic serving platform that leverages workflow awareness and stage isolation to improve scheduling and KV cache efficiency in agentic NL2SQL workflows.

**Brown University** Sep 2023 – May 2024  
*Student Researcher* Providence, RI, US  
Advisor: Prof. [Nikos Vasilakis](#)  
Contributed to [PASH](#), an open-source [Linux Foundation](#) project. Worked on improving the performance of shell scripts by running them on serverless infrastructure, and providing fault tolerance for distributed shell script execution.

**Arrikto Inc.** Dec 2022 – Mar 2023  
*Software Engineering Intern* Athens, Greece

Supervisor: [Vangelis Koukis](#)

Explored how to serve PyTorch models more efficiently through KServe, a highly scalable machine learning deployment platform for Kubernetes.

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| Awards       | <a href="#">Columbia University Presidential Fellowship</a>                                                                                                                                                                                                                                                           | 2024 – Present                |
|              | A.G. Leventis Foundation Scholarship                                                                                                                                                                                                                                                                                  | 2025 – Present                |
|              | Gerondelis Foundation Scholarship                                                                                                                                                                                                                                                                                     | 2025                          |
|              | ACM SOSP 2025 Student Travel Grant                                                                                                                                                                                                                                                                                    | 2025                          |
| Publications | <a href="#">VineLM: Trie-Based Fine-Grained Control for Agentic Workflows</a><br>Nikos Pagonas, Matthew Lou, Tianyi Peng, Dan Rubenstein, Kostis Kaffes                                                                                                                                                               |                               |
|              | <a href="#">Agentic Data Environments</a><br>Elaine Ang, Chenxi Huang, Georgios Liargkovas, Jerry Liu, Jinhui Liu, Nikos Pagonas, Charlie Summers, Haonan Wang, Jiakai Xu, Tianle Zhou, Yusen Zhang, Zhou Yu, Zhuo Zhang, Tianyi Peng, Kostis Kaffes, Eugene Wu<br><i>IEEE Data Engineering Bulletin</i> , March 2026 |                               |
|              | <a href="#">AgentOpt v0.1 Technical Report: Client-Side Optimization for LLM-Based Agents</a><br>Wenyue Hua, Sripad Karne, Qian Xie, Armaan Agrawal, Nikos Pagonas, Kostis Kaffes, Tianyi Peng                                                                                                                        |                               |
|              | <a href="#">Cortex: Workflow-Aware Resource Pooling and Scheduling for Agentic Serving</a><br>Nikos Pagonas, Yeounoh Chung, Kostis Kaffes, Arvind Krishnamurthy<br><i>SOSP 2025 Workshop on Systems for Agentic AI (SAA 2025)</i>                                                                                     |                               |
|              | <b>Which Model for Which Role? Sample-Efficient Agent Configuration Search</b><br><i>(Under Submission)</i>                                                                                                                                                                                                           |                               |
|              | <b><math>\lambda</math>easeh: Scaling Out Shell Scripts with Reliable Cloud Computing</b><br><i>(Under Submission)</i>                                                                                                                                                                                                |                               |
| Service      | <b>Artifact Evaluation Committees</b><br><a href="#">EuroSys 2027</a> , <a href="#">SOSP 2026</a> , <a href="#">CAIS 2026</a> , <a href="#">EuroSys 2026</a> , <a href="#">SOSP 2025</a> , <a href="#">OSDI 2025</a> , <a href="#">ATC 2025</a>                                                                       |                               |
|              | <b>Web and Publication Chair</b><br><a href="#">EuroSec 2024</a>                                                                                                                                                                                                                                                      |                               |
| Teaching     | <b>Columbia University</b><br><i>Teaching Assistant</i><br>Course: Networks, Crowds, and the Web<br>Professor: <a href="#">Augustin Chaintreau</a>                                                                                                                                                                    | Fall 2025<br>New York, NY, US |

**Columbia University** Spring 2025  
*Teaching Assistant* New York, NY, US  
Course: [Agentic Systems Made Real](#)  
Professors: [Eugene Wu](#), [Kostis Kaffes](#)

**Brown University** Fall 2024  
*Teaching Assistant* Providence, RI, US  
Course: [Systems Transforming Systems](#)  
Professor: [Nikos Vasilakis](#)  
Guided students through a systems research project, including implementation, evaluation, and paper writing. Paper submitted to USENIX ATC 2025.

**Brown University** Spring 2024  
*Teaching Assistant* Providence, RI, US  
Course: [Distributed Systems](#)  
Professor: [Nikos Vasilakis](#)  
Helped revamp the course by redesigning its semester-long project and implementing its autograding infrastructure from scratch.

**National Technical University of Athens** Spring 2022  
*Lab Assistant* Athens, Greece  
Course: Operating Systems  
Professors: [N. Koziris](#), [G. Goumas](#)

**National Technical University of Athens** Spring 2020  
*Lab Assistant* Athens, Greece  
Course: Programming Techniques  
Professors: [N. Papaspyrou](#), [A. Pagourtzis](#), [G. Stamou](#), [G. Goumas](#)

**National Technical University of Athens** Fall 2019  
*Lab Assistant* Athens, Greece  
Course: Introduction to Programming  
Professors: [N. Papaspyrou](#), [D. Fotakis](#), [S. Zachos](#)

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| <b>Research<br/>Mentoring</b> | <a href="#">Armaan Agrawal</a> May 2026 – Present<br><i>Columbia University, BS</i>                            |
|                               | <a href="#">Grace Li</a> Sep 2025 – Present<br><i>Columbia University, BS</i>                                  |
|                               | <a href="#">Matthew Lou</a> Sep 2025 – Jun 2026<br><i>Columbia University, BS → next: Hudson River Trading</i> |
|                               | <a href="#">Joey Huang</a> Sep 2025 – Dec 2025<br><i>Columbia University, BS → next: Google</i>                |

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|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|----------|
| Invited Talks | VineLM: Trie-Based Fine-Grained Control for Agentic Workflows<br><i>Languages, Systems, and Data Seminar @ University of California, Santa Cruz</i><br>Host: Lindsey Kuper |                                                     | May 2025 |
| Skills        | Prog. Languages                                                                                                                                                            | Python, C, Shell, Javascript, Java, Go              |          |
|               | Frameworks                                                                                                                                                                 | LangChain, LangGraph                                |          |
|               | Platforms                                                                                                                                                                  | AWS, Docker, Kubernetes, Apache Kafka, Apache Spark |          |
|               | Tools                                                                                                                                                                      | Git, LaTeX                                          |          |
| Languages     | English                                                                                                                                                                    | Fluent — TOEFL iBT 116/120 (C2)                     |          |
|               | Greek                                                                                                                                                                      | Native                                              |          |